Ansible - adhoc commands and playbook

What are Ad-Hoc Commands in Ansible?

An ad-hoc command is a one-line instruction used to perform a quick task on remote machines without writing a playbook.

It is imperative automation, not structured automation.

These are the example of ad hoc commands.

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m copy -a "src=hari dest=/tmp"
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "checksum": "42ac8fad543cfa32fe27fea08a62beb5b9e299a8",
    "dest": "/tmp/hari",
    "gid": 1000,
    "group": "ec2-user",
    "md5sum": "fc93cab59237d57daf8cd657a540fbb1",
    "mode": "0664",
    "owner": "ec2-user",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 13,
    "src": "/home/ec2-user/.ansible/tmp/ansible-tmp-1771433383.2450376-3612-252473401499282/source",
    "state": "file",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "checksum": "42ac8fad543cfa32fe27fea08a62beb5b9e299a8",
    "dest": "/tmp/hari",
    "gid": 1000,
    "group": "ec2-user",
    "md5sum": "fc93cab59237d57daf8cd657a540fbb1",
    "mode": "0664",
    "owner": "ec2-user",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 13,
    "src": "/home/ec2-user/.ansible/tmp/ansible-tmp-1771433383.2159321-3611-163560848913361/source",
    "state": "file",
    "uid": 1000
}
[ec2-user@ip-172-31-89-110 ~]$
```

```
Instances | EC2 | us-east-1
EC2 Instance Connect | us-east
Ansible - Ad hoc Commands -
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0bbfe72464f7c7c11&osUser=ec2-user&region=us-eas...
aws  Search [Alt+S]  United States (N. Virginia)  Dhanam M (0517-4104-1276)  Dhanam M

       __|  __|_  )
       _|  (     /   Amazon Linux 2023
      ___|\___|___|
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Wed Feb 18 16:49:44 2026 from 172.31.89.110
[ec2-user@ip-172-31-28-3 ~]$ ls
[ec2-user@ip-172-31-28-3 ~]$ cd/tmp
-bash: cd/tmp: No such file or directory
[ec2-user@ip-172-31-28-3 ~]$ cd /tmp
[ec2-user@ip-172-31-28-3 tmp]$ ls
hari
systemd-private-db2a7c08772d42a381822a2943f302fd-chronyd.service-CmViTE
systemd-private-db2a7c08772d42a381822a2943f302fd-dbus-broker.service-Mj295G
systemd-private-db2a7c08772d42a381822a2943f302fd-policy-routes@ens5.service-8b52df
systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-logind.service-eslAuf
systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-resolved.service-NGfy92
[ec2-user@ip-172-31-28-3 tmp]$

i-0bbfe72464f7c7c11 (ansible-slave)
PublicIPs: 52.73.204.225  PrivateIPs: 172.31.28.3

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```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m command -a "uptime"
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED | rc=0 >>
 16:45:29 up 16 min,  1 user,  load average: 0.08, 0.03, 0.01
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED | rc=0 >>
 16:45:29 up 16 min,  1 user,  load average: 0.00, 0.00, 0.00
```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m ping
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m setup
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | SUCCESS => {
    "ansible_facts": {
        "ansible_all_ipv4_addresses": [
            "172.31.27.165"
        ],
        "ansible_all_ipv6_addresses": [
            "fe80::8ff:e4ff:fe76:2a09"
        ],
        "ansible_apparmor": {
            "status": "disabled"
        },
        "ansible_architecture": "x86_64",
        "ansible_bios_date": "10/16/2017",
        "ansible_bios_vendor": "Amazon EC2",
        "ansible_bios_version": "1.0",
        "ansible_board_asset_tag": "i-0bb8079c314c6de2e",
        "ansible_board_name": "NA",
        "ansible_board_serial": "NA",
```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m file -a "path=/tmp/testdir mode=0755 state=directory"
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "gid": 1000,
    "group": "ec2-user",
    "mode": "0755",
    "owner": "ec2-user",
    "path": "/tmp/testdir",
    "secontext": "unconfined_u:object_r:user_tmp_t:s0",
    "size": 40,
    "state": "directory",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "gid": 1000,
    "group": "ec2-user",
    "mode": "0755",
    "owner": "ec2-user",
    "path": "/tmp/testdir",
    "secontext": "unconfined_u:object_r:user_tmp_t:s0",
    "size": 40,
    "state": "directory",
    "uid": 1000
}
[ec2-user@ip-172-31-89-110 ~]$
```

```
[ec2-user@ip-172-31-28-3 tmp]$ ls
hari                                                                systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-logind.service-eslAuf
systemd-private-db2a7c08772d42a381822a2943f302fd-chronyd.service-CmViTE   systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-resolved.service-NGfy92
systemd-private-db2a7c08772d42a381822a2943f302fd-dbus-broker.service-Mj295G   testdir
systemd-private-db2a7c08772d42a381822a2943f302fd-policy-routes@ens5.service-8b52df
[ec2-user@ip-172-31-28-3 tmp]$
```

i-0bbfe72464f7c7c11 (ansible-slave)

PublicIPs: 52.73.204.225 PrivateIPs: 172.31.28.3

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What Is a Playbook?

A playbook in Ansible is a YAML file that defines automation in a structured, repeatable way.

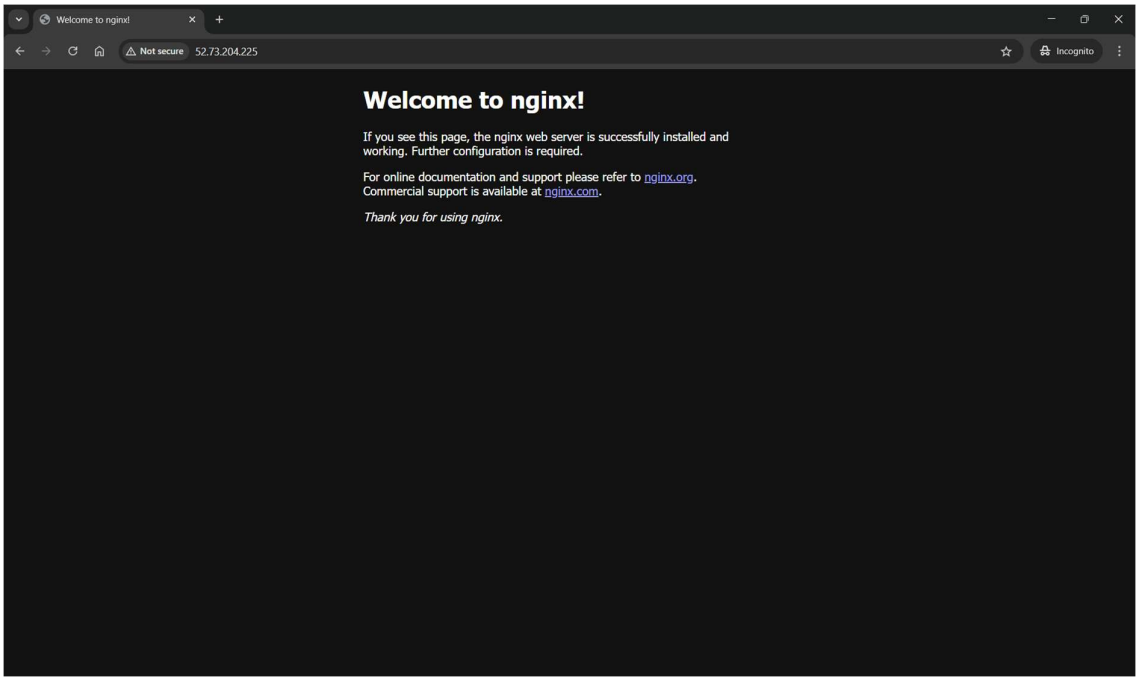
It describes:

- Which machines to target

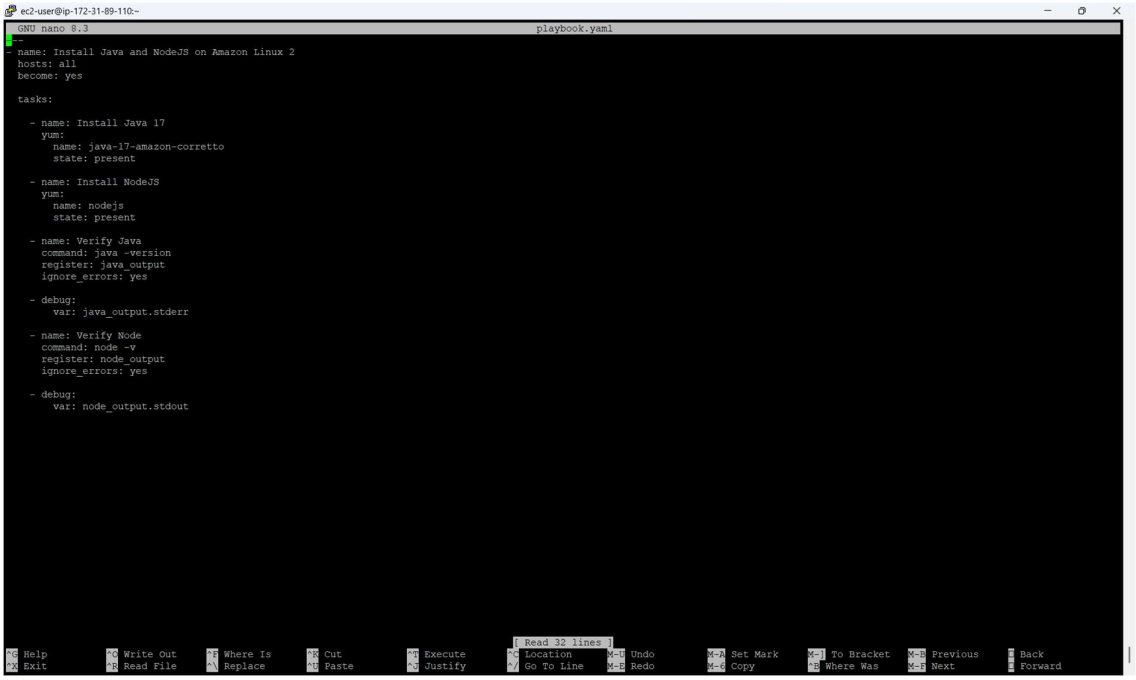
- What tasks to run
- In what order
- With what privileges
- Under what conditions

```
---
- name: Install and Start Nginx
  hosts: webservers
  become: yes
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
    - name: Start nginx
      service:
        name: nginx
        state: started
```

I have installed the nginx and hosted on the web.



And then I have installed the JAVA , NODE.js .



The process have been done using the playbook.

ec2-user@ip-172-31-89-110:~

```
      ignore_errors: yes

    - debug:
        var: node_output.stdout
[ec2-user@ip-172-31-89-110 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [Install Java and NodeJS on Amazon Linux 2] ******************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.28.3]
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.27.165]

TASK [Install Java 17] *********************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [Install NodeJS] **********************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [Verify Java] *************************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [debug] *******************************************************************************************************************************************************
ok: [172.31.28.3] => {
    "java_output.stderr": "openjdk version \"17.0.18\" 2026-01-20 LTS\nOpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)\nOpenJDK 64-Bit Server VM Corretto-17.0.18.9.1
(build 17.0.18+9-LTS, mixed mode, sharing)"
}
ok: [172.31.27.165] => {
    "java_output.stderr": "openjdk version \"17.0.18\" 2026-01-20 LTS\nOpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)\nOpenJDK 64-Bit Server VM Corretto-17.0.18.9.1
(build 17.0.18+9-LTS, mixed mode, sharing)"
}

TASK [Verify Node] *************************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [debug] *******************************************************************************************************************************************************
ok: [172.31.28.3] => {
    "node_output.stdout": "v18.20.8"
}
ok: [172.31.27.165] => {
    "node_output.stdout": "v18.20.8"
}

PLAY RECAP *********************************************************************************************************************************************************
172.31.27.165              : ok=7    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.28.3                : ok=7    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-89-110 ~]$
```